



MSA

October University for Modern Science & Arts

Faculty of Engineering

Department of Mechatronics Systems Engineering

Machine Design (MSE243)

Designing a Speed Increaser Gearbox

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Design Requirements and Specifications

The project task is to design a 3-stage of step-up gearbox comprised of a total of 4 shafts, 2 mating spur gears, 2 mating helical gears, and 2 connected belt pulleys. The specific parameters to operate on are:

- ❖ Input Power: 3.5 HP = 2.611 kW
- ❖ Input Speed: 750 rpm
- ❖ Output Speed: 1600 rpm
- ❖ Number of Stages: 3
- ❖ First Stage: Helical Gear
- ❖ Second Stage: Spur Gear
- ❖ Third Stage: Belt

Design Calculations: Gears

3.1 Helical Gears

- ❖ Input Speed: 750 rpm
- ❖ Output Speed: 970 rpm
- ❖ Reduction Ratio: 1.27
- ❖ $P_t = \pi * m = 8.63$
- ❖ Pressure angle $\alpha =$ assume 20
- ❖ $P_x = \frac{P_t}{\tan \alpha} = (8.63/\tan 20) = 23.71$
- ❖ $F = 2P_x m \approx 59.275 \text{ mm}$
- ❖ $F_a = w_t \tan \beta$
- ❖ Helix angle $\beta =$ assume 20
- ❖ $N_1 = 18$ teeth
- ❖ $m = 2.54 \rightarrow$ Choose 2.75 (Next choice)
- ❖ $d_1 = N_1 * m = 18 * 2.75 = 49.5 \text{ mm}$
- ❖ $N_2 = N_1 * RR = 18 * 1.29 \approx 24$ teeth
- ❖ $d_2 = N_2 * m = 24 * 2.75 = 66 \text{ mm}$

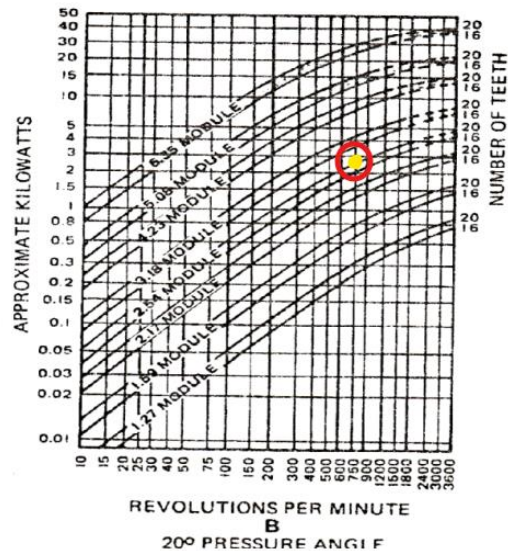


Figure 1 - Module Table

Safety Checks:

1. Bending Stress:

$$w_t = \frac{\text{Power}}{\text{Velocity}} = \frac{2611}{\left(\frac{\pi * D * N}{60}\right)} = \frac{2611}{\left(\frac{\pi * 0.0495 * 750}{60}\right)} = 1340 \text{ N}$$

$$K_v = \frac{6}{6 + V} = 0.7556, Y = 0.29327$$

$$\sigma = \frac{w_t}{F * K_v * m * Y}$$

Number of teeth	$\phi = 20^\circ$ $a = 0.8^*$ $b = m$	$\phi = 20^\circ$ $a = m$ $b = 1.25m$	$\phi = 25^\circ$ $a = m$ $b = 1.25m$	$\phi = 25^\circ$ $a = m$ $b = 1.35m^+$
12	0.335 12	0.229 60	0.276 77	0.254 73
13	0.348 27	0.243 17	0.292 81	0.271 77
14	0.359 85	0.255 30	0.307 17	0.287 11
15	0.370 13	0.266 22	0.320 09	0.301 00
16	0.379 31	0.276 10	0.331 78	0.313 63
17	0.387 57	0.285 08	0.342 40	0.325 17
18	0.395 79	0.293 79	0.352 81	0.335 74
19	0.401 75	0.300 78	0.360 99	0.345 46
20	0.407 97	0.307 69	0.369 16	0.354 44

Figure 2 - Lewis Factor

$$\sigma = \frac{1340}{0.059275 * 0.7556 * 2.75 * 0.29327} = 37.097 \text{ MPa}$$

$$\sigma_{design} = F.S * \sigma = 2 * 37.097 = 74.19 \text{ MPa}$$

Choose 1095 As rolled ; $\sigma_y = 572.3 \text{ MPa}$, $H_B = 293$

2. Bending Fatigue:

$$S_e = K_a K_b K_c K_d K_e K_f S_e'$$

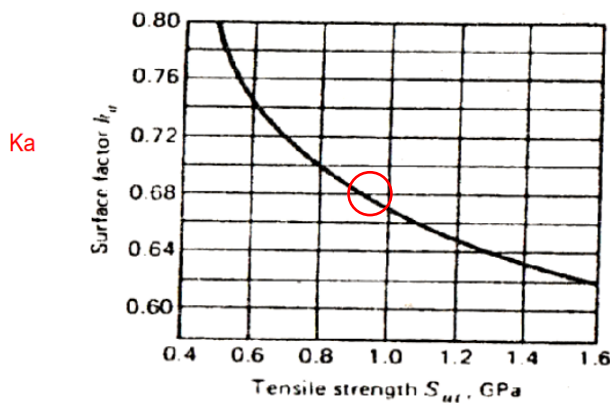


Figure 3 - Ka Table

Module m	Factor Kb	Module m	Factor Kb
1 to 2	1.000	11	0.843
2.25	0.984	12	0.836
2.5	0.974	14	0.824
2.75	0.965	16	0.813
3	0.956	18	0.804
3.5	0.942	20	0.796
4	0.930	22	0.788
4.5	0.920	25	0.779
5	0.910	28	0.770
5.5	0.902	32	0.760
6	0.894	36	0.752
7	0.881	40	0.744
8	0.870	45	0.736
9	0.860	50	0.728
10	0.851		

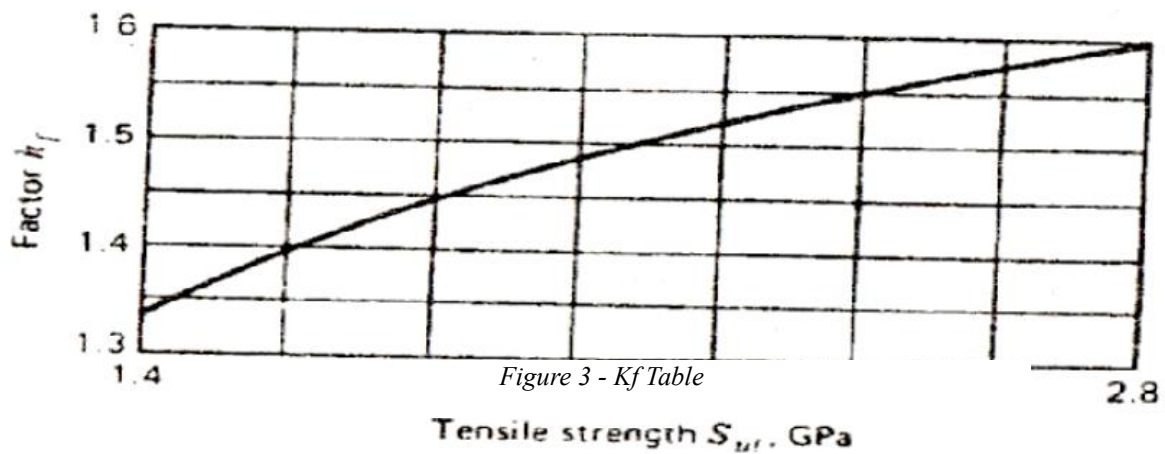
Figure 4 - Kb Table

Reliability R	0.50	0.90	0.95	0.99	0.999	0.9999
Factor Kc	1.000	0.897	0.868	0.814	0.753	0.720

K_d = Temperature Factor

$$= \begin{cases} 1 & T \leq 350 \\ 0.5 & 350 < T \leq 500 \end{cases}$$

$K_e = \text{Modified factor for stress concentration} = 1$



$$K_a = 0.68$$

$$K_b = 0.956$$

$$K_c = 0.725$$

$$K_d = 1$$

$$K_e = 1$$

$$K_f = 1$$

Number Of teeth	Number of teeth in mating gear						
	1	17	25	35	50	85	300
18	0.244 86	0.324 04	0.332 14	0.338 40	0.344 04	0.350 50	0.355 94
19	0.247 94	0.330 29	0.338 78	0.345 37	0.351 34	0.358 22	0.364 05
20	0.250 72	0.336 00	0.344 85	0.351 76	0.385 04	0.365 32	0.371 51

Figure 4 - Geometry Factor

$$S_e' = \frac{\sigma_{ult}}{2} = \frac{955.3}{2} = 482.65 \text{ MPa}$$

$$S_e = 229.61 \text{ MPa}$$

$$\sigma_a = \frac{1340}{0.059 * 0.7556 * 2.75 * 0.33214} = 32.755 \text{ MPa}$$

$$n_G = \frac{S_e}{\sigma_a} = \frac{2291.61}{32.755} = 7$$

$$n = \frac{n_G}{K_o K_m} = \frac{7}{1 * 1.3} = 4.3 > 1$$

Safe

3. Surface Durability:

$$S_c = 2.76 * H_B - 70 = 2.76 * 293 - 70 = 738.68 \text{ MPa}$$

$$S_H = \frac{C_L C_H}{C_T C_R} * S_c = \frac{1 * 1}{1 * 1.25} * 738.68 = 590.944 \text{ MPa}$$

$$\sigma_H = 590.944 \text{ MPa} = C_p \sqrt{\frac{w_{tp}}{C_v * F * d * I}}$$

$$I = \frac{\cos \phi \sin \phi}{2} * \frac{m}{m + 1} = 0.1178$$

$$\sigma_H = 191 \sqrt{\frac{w_{tp}}{0.7556 * 0.059 * 0.0495 * 0.1178}}$$

$$w_{tp} = 2492 \text{ N}$$

$$n_G = \frac{w_{tp}}{w_t} = \frac{2492}{1340} = 1.86$$

$$n = \frac{n_G}{c_o c_m} = \frac{1.86}{1 * 1.3} = 1.14 > 1$$

Safe

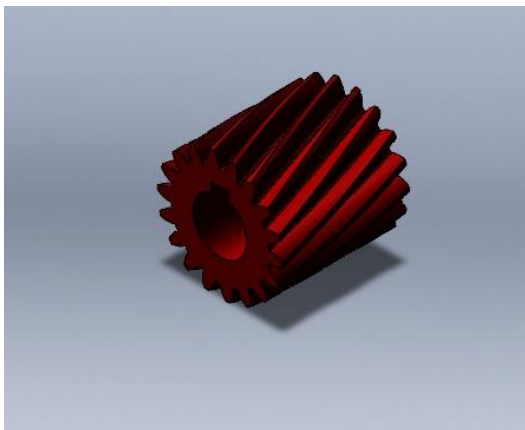


Figure 6 - Pinion Helical Gear

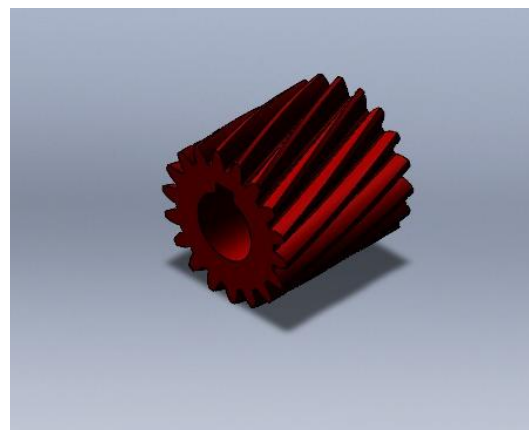


Figure 7 - Driven Helical Gear

3.2 Spur Gears

- ❖ Input Speed: 750 rpm
- ❖ Output Speed: 970 rpm
- ❖ Reduction Ratio: 1.3
- ❖ $N_1 = 18$ teeth
- ❖ $m = 2.75$
- ❖ $d_1 = N_1 * m = 18 * 2.75 = 49.5$ mm
- ❖ $N_2 = N_1 * RR = 18 * 1.3 \approx 24$ teeth
- ❖ $d_2 = N_2 * m = 24 * 2.75 = 66$ mm
- ❖ $F = 4 * \pi * m = 34.55 \text{ mm} \approx 35 \text{ mm}$

Safety Checks:

1- Bending Stress:

$$w_t = \frac{\text{Power}}{\text{Velocity}} = \frac{2611}{\left(\frac{\pi * D * N}{60}\right)} = \frac{2611}{\left(\frac{\pi * 0.0495 * 970}{60}\right)} = 1028 \text{ N}$$

$$K_v = \frac{6}{6 + V} = 0.7, Y = 0.29327$$

$$\sigma = \frac{w_t}{F * K_v * m * Y} = \frac{1038.55}{0.035 * 0.7 * 2.75 * 0.29327} = 52 \text{ MPa}$$

$$\sigma_{design} = F.S * \sigma = 2 * 52 = 104 \text{ MPa} < \sigma_y$$

Safe

2- Bending Fatigue

$$S_e = K_a K_b K_c K_d K_e K_f S_e'$$

$$K_a = 0.8$$

$$K_b = 0.965$$

$$K_c = 0.725$$

$$K_d = 1$$

$$K_e = 1$$

$$K_f = 1$$

$$S_e' = \frac{\sigma_{ult}}{2} = \frac{424}{2} = 212 \text{ MPa}$$

$$S_e = 118.65 \text{ MPa}$$

$$\sigma_a = \frac{w_t}{F * K_v * m * J} = \frac{1038.55}{0.035 * 0.88 * 2.75 * 0.33814} = 36.54 \text{ MPa}$$

$$n_G = \frac{S_e}{\sigma_a} = \frac{118.65}{36.54} = 3.255$$

$$n = \frac{n_G}{K_o K_m} = \frac{3.255}{1 * 1.3} = 2.5 > 1$$

Safe

3- Surface Durability:

$$S_c = 263.96 \text{ MPa}$$

$$S_H = \frac{C_L C_H}{C_T C_R} * S_c = \frac{1 * 1}{1 * 1.3} * 263.96 = 343.148 \text{ MPa}$$

$$\sigma_H = C_p \sqrt{\frac{w_t}{C_v * F * d * I}} ; I = \frac{\cos \phi \sin \phi}{2} * \frac{m}{m + 1} = 0.1175$$

$$\sigma_H = 191 \sqrt{\frac{584}{0.88 * 0.035 * 0.05 * 0.1175}} = 343.148 \text{ MPa}$$

$$n_G = \frac{S_H}{\sigma_H} = \frac{584}{1038} = 0.568$$

$$n = \frac{n_G}{c_o c_m} = \frac{0.568}{1 * 1.3} = 0.437 < 1$$

Not Safe

Changed material to 1050 Normalized ;

$$H_B = 201$$

$$S_c = 484.76 \text{ MPa}$$

$$S_H = \frac{C_L C_H}{C_T C_R} * S_c = \frac{1 * 1}{1 * 1.3} * 484.76 = 630.18 \text{ MPa}$$

$$w_{tp} = 1.915 \text{ N}$$

$$n = \frac{n_G}{c_o c_m} = \frac{1.915}{1 * 1.3} = 1.473 > 1$$

Safe

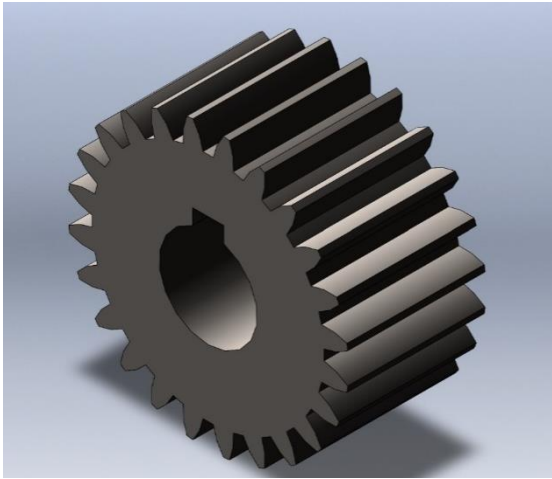


Figure 6 - Pinion Spur Gear

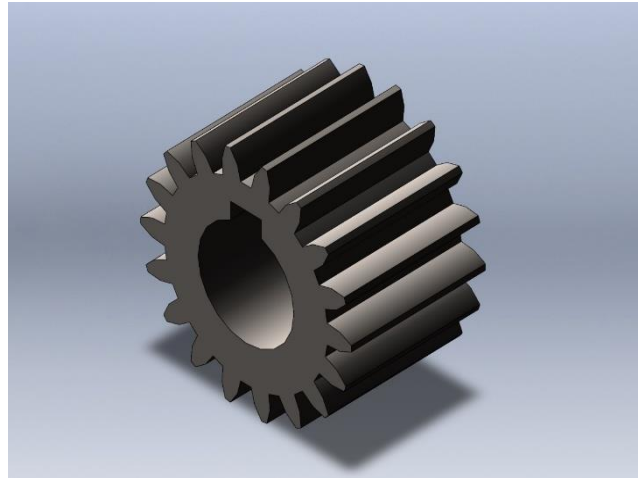


Figure 5 - Driven Spur Gear

Pulley and Belt Design

Input Power= 3.5 Hp = 2.611 KW

❖ Input Speed = 1260 rpm

❖ Reduction ratio= 1.27

(a) Speed Ratio

Speed ratio=0.02

(b) Service Factor

From Table 3 the service factor is 1.4 .as we use an internal combustion engine with 4 cylinders

➔ Class 2➔ Agitators and Mixers ➔ (over 16) hours

SPEED INCREASE RATIO		TYPES OF PRIME MOVER					
		'Soft' starts			'Heavy' starts		
For speed increasing drives of: Speed ratio 1.00 – 1.24 multiply service factor by 1.00 Speed ratio 1.25 – 1.74 multiply service factor by 1.05 Speed ratio 1.75 – 2.49 multiply service factor by 1.11 Speed ratio 2.50 – 3.49 multiply service factor by 1.18 Speed ratio 3.50 and over multiply service factor by 1.25		Electric motors: AC – Star Delta start DC – Shunt wound Internal combustion engines with 4 or more cylinders Prime movers fitted with centrifugal clutches, dry or fluid couplings or electronic soft start devices			Electric motors: AC – Direct-on-line start DC – Series & compound wound Internal combustion engines with less than 4 cylinders Prime movers not fitted with soft start devices		
		Hours per day duty					
	TYPES OF DRIVEN MACHINE	10 and under	Over 10 to 16	Over 16	10 and under	Over 10 to 16	Over 16
Class 1 Light Duty	Agitators (uniform density), blowers, exhausters and fans (up to 75 kW), centrifugal compressors and pumps. Belt conveyors (uniformly loaded).	1.0	1.1	1.2	1.1	1.2	1.3
Class 2 Medium Duty	Agitators and mixers (variable density), blowers, exhausters and fans (over 75 kW). Rotary compressors and pumps (other than centrifugal). Belt conveyors (not uniformly loaded), generators and exciters, laundry machinery, lineshafts, machine tools, printing machinery, sawmill and woodworking machinery, screens (rotary).	1.1	1.2	1.3	1.2	1.3	1.4
Class 3 Heavy Duty	Brick machinery, bucket elevators, compressors and pumps (reciprocating), conveyors (heavy duty). Hoists, mills (hammer), pulverisers, punches, presses, shears, quarry plant, rubber machinery, screens (vibrating), textile machinery.	1.2	1.3	1.4	1.4	1.5	1.6
Class 4 Extra Heavy Duty	Crushers (gyratory-jaw roll), mills (ball-rod-tube).	1.3	1.4	1.5	1.5	1.6	1.8

(c) Design Power

$$DP = 2.611 \times 1.4 = 3.6554 \text{ KW}$$

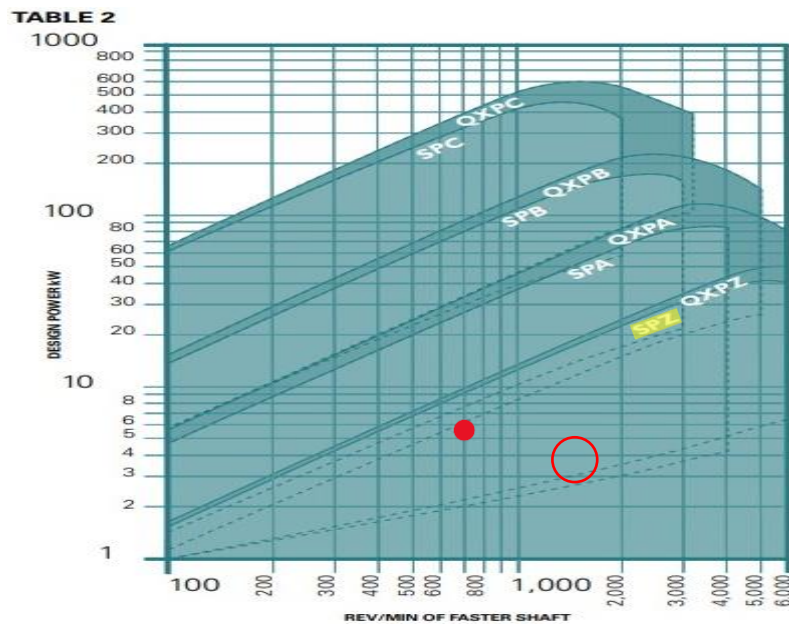


Figure 7 - Belt Type Selection

(d) Belt Type

From figure 10, the intersection of design power (3.6554 KW) and the speed of the faster shaft (1600rpm) indicate **SPZ** section.

(e) Minimum Pulley

TABLE 1

Speed of faster shaft rev/min	*Minimum Pulley Diameter (mm)																			
	Design Power (kW)																			
	up to 1	3.0	4.0	5.0	7.5	10	15	20	25	30	40	50	60	75	90	110	130	150	200	250
500	56	90	100	112	125	140	180	200	212	236	250	280	280	315	375	400	450	475	500	560
600	56	85	90	100	112	125	140	180	200	212	224	250	265	280	300	335	375	400	475	500
720	56	80	85	90	100	106	132	150	160	170	200	236	250	265	280	300	335	375	450	500
960	56	75	80	85	95	100	112	132	150	180	180	200	224	250	280	280	300	335	400	450
1200	56	71	80	80	95	95	106	118	132	150	160	180	200	236	236	250	265	300	335	355
1440	56	63	75	80	85	85	100	112	125	140	160	170	190	212	236	236	250	280	315	335
1800	56	63	71	75	80	85	95	106	112	125	150	160	170	190	212	224	236	265	300	335
2880	56	60	67	67	80	80	85	90	100	112	125	140	160	170	180	212	224	236	—	—

From Table 1 (page 38) using **interpolation** the minimum recommended pulley is approximately **70 mm**.

(f) Pulley Pitch Diameters

By referring to the center distance tables relating to SPZ section belts on page 47 at reduction ratio 1.293, the selection is a driving pulley of **70 mm** pitch diameter and a driven pulley of **71 mm** and **90 mm** pitch diameter.



Combined Arc and Belt Length Correction Factor				0.80		0.85		0.90		0.95		1.00		1.05		1.10		1.15					
Speed Ratio	Pitch Diameter of Pulleys		Power per SPZ Belt (kW)		BELT LENGTH																SPZ & OXPZ ONLY		Speed Ratio
	Driver	Driven	1440 rev/min	2880 rev/min	630	710	800	900	1000	1120	1250	1400	1600	1800	2000	2240	2500	2800	3150	3550			
2.09	67	140	1.46	2.54	148	189	235	285	335	396	461	536	636	737	837	957	1087	1237	1412	1612	2.09		
2.11	95	200	2.70	4.81	—	—	—	212	263	324	390	465	566	666	767	887	1017	1167	1342	1542	2.11		
2.12	85	180	2.26	4.02	—	—	186	237	288	349	414	490	590	690	790	911	1041	1191	1366	1566	2.12		
2.13	75	160	1.82	3.21	—	165	211	262	313	373	438	514	614	714	814	934	1065	1215	1390	1590	2.13		
2.22	63	140	1.28	2.21	151	192	237	288	338	399	464	539	639	740	840	960	1090	1240	1415	1615	2.22		
2.22	90	200	2.48	4.42	—	—	—	215	267	328	393	469	570	670	770	891	1021	1171	1346	1546	2.22		
2.23	56	125	1.15	2.00	169	210	256	306	356	416	—	—	—	—	—	—	—	—	—	—	2.23		
2.23	112	250	3.43	6.12	—	—	—	—	204	267	334	410	511	612	712	833	963	1114	1289	1489	2.23		
2.25	80	180	2.04	3.61	—	—	189	241	292	352	418	493	594	694	794	914	1045	1195	1370	1570	2.25		
2.25	140	315	4.61	8.16	—	—	—	—	—	—	252	331	434	535	637	758	888	1039	1214	1415	2.25		

(g) Belt Length, Centre Distance and Correction Factor

Reading across the table, the nearest center distance to the 873 mm required is 994 **mm**. The belt length at the head of the column is **2240 mm**. By following the shading of the color to the top of the table, a correction factor of **1.05** is noted.

(h) Rated Power per Belt

From the power rating table (pages 54 & 57), the rated power is **1.57kW** (Using interpolation) for Power Plus SPZ (power per belt).

Rev/min of faster shaft	RATED POWER (kW) PER BELT FOR SMALL PULLEY PITCH DIA (mm)												Belt Speed (m/s)
	63	67	71	75	80	85	90	95	100	112	125	140	
100	0.12	0.13	0.15	0.16	0.18	0.20	0.22	0.24	0.25	0.30	0.35	0.40	10
500	0.45	0.52	0.59	0.66	0.74	0.82	0.91	0.99	1.07	1.27	1.48	1.73	
720	0.61	0.70	0.80	0.89	1.01	1.13	1.25	1.36	1.48	1.76	2.05	2.39	
800	0.66	0.77	0.87	0.98	1.11	1.24	1.37	1.50	1.62	1.93	2.25	2.63	
900	0.73	0.85	0.96	1.08	1.23	1.37	1.51	1.66	1.80	2.14	2.50	2.92	
960	0.77	0.89	1.02	1.14	1.29	1.45	1.60	1.75	1.90	2.26	2.65	3.09	
1000	0.79	0.92	1.05	1.18	1.34	1.50	1.66	1.81	1.97	2.34	2.74	3.20	
1100	0.85	1.00	1.14	1.28	1.45	1.63	1.80	1.97	2.14	2.55	2.98	3.48	
1200	0.92	1.07	1.22	1.37	1.56	1.75	1.94	2.12	2.31	2.75	3.22	3.75	
1300	0.98	1.14	1.30	1.47	1.67	1.87	2.07	2.27	2.47	2.94	3.45	4.02	
1400	1.03	1.21	1.39	1.56	1.78	1.99	2.21	2.42	2.63	3.14	3.68	4.29	
1440	1.06	1.24	1.42	1.60	1.82	2.04	2.26	2.48	2.70	3.22	3.77	4.40	
1500	1.09	1.28	1.47	1.65	1.88	2.11	2.34	2.57	2.79	3.33	3.90	4.55	
1600	1.15	1.35	1.54	1.74	1.99	2.23	2.47	2.71	2.95	3.52	4.12	4.81	
1700	1.20	1.41	1.62	1.83	2.09	2.34	2.60	2.85	3.11	3.70	4.34	5.06	

(j) Additional Power Increment

The power increment for a speed ratio of 2.38 is **0.02kW**(Using interpolation). (pages 54)

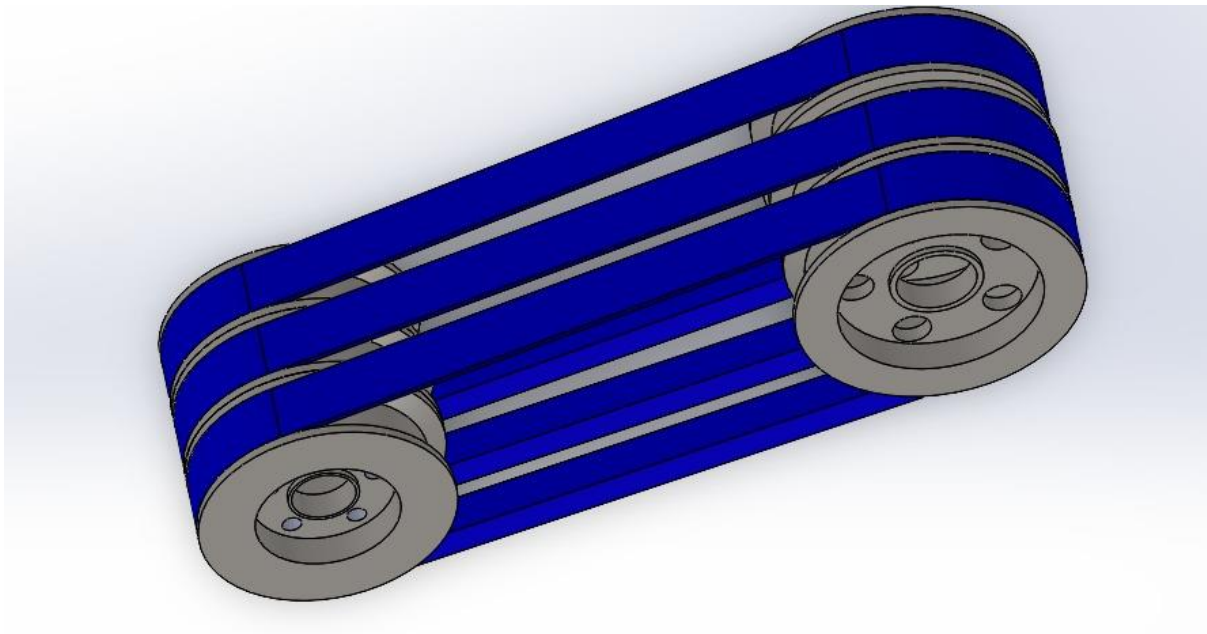
Rev/min of faster shaft	ADDITIONAL POWER (kW) PER BELT FOR SPEED RATIO									
	1.00 to 1.01	1.02 to 1.05	1.06 to 1.11	1.12 to 1.18	1.19 to 1.26	1.27 to 1.38	1.39 to 1.57	1.58 to 1.94	1.95 to 3.38	3.39 and over
100	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01	0.02	0.02
500	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
720	0.00	0.01	0.03	0.05	0.06	0.08	0.09	0.10	0.11	0.12
800	0.00	0.01	0.03	0.05	0.07	0.08	0.10	0.11	0.12	0.13
900	0.00	0.01	0.03	0.06	0.08	0.10	0.11	0.13	0.14	0.14
960	0.00	0.01	0.04	0.06	0.08	0.10	0.12	0.13	0.15	0.15
1000	0.00	0.01	0.04	0.06	0.09	0.11	0.12	0.14	0.15	0.16
1100	0.00	0.01	0.04	0.07	0.10	0.12	0.14	0.15	0.17	0.18
1200	0.00	0.02	0.04	0.08	0.10	0.13	0.15	0.17	0.18	0.19
1300	0.00	0.02	0.05	0.08	0.11	0.14	0.16	0.18	0.20	0.21
1400	0.00	0.02	0.05	0.09	0.12	0.15	0.17	0.20	0.21	0.23
1440	0.00	0.02	0.05	0.09	0.13	0.15	0.18	0.20	0.22	0.23
1500	0.00	0.02	0.05	0.10	0.13	0.16	0.19	0.21	0.23	0.24
1600	0.00	0.02	0.06	0.10	0.14	0.17	0.20	0.22	0.24	0.26
1700	0.00	0.02	0.06	0.11	0.15	0.18	0.21	0.24	0.26	0.27

(k) Corrected Power per Belt

$$CP = (0.02 + 1.57) \times 1.05 = \mathbf{1.67 \text{ kW}} \text{ per belt.}$$

(l) Number of Belts required

$$\text{No. of belts} = \frac{3.6554}{1.67} = 2.188 \rightarrow \mathbf{3 \text{ SPZ Power Plus wedge belts.}}$$



Therefore, the results of the belt design are the following:

- ❖ Input Speed: 1260 rpm
- ❖ Output Speed: 1600 rpm
- ❖ Reduction Ratio: 1.27
- ❖ Power: 2611 W
- ❖ Service Factor = 1.4
- ❖ Design Power =
 $2.611 \times 1.4 = 3.6554 \text{ kW}$
- ❖ Belt Type: SPZ
- ❖ Minimum Pulley
Diameter = 70 mm
- ❖ Pulley Diameters = 71 & 90 mm
- ❖ Center Distance = 873-994 mm
- ❖ Belt Length = 2000-2240 mm
- ❖ Correction Factor = 1.05
- ❖ Rated Power = 1.57 kW
- ❖ Additional Power
Increment = 0.02 kW per belt
- ❖ Power Per Belt =
 $1.05(1.57 + 0.02) =$
1.67 kW/belt

$$\text{Number of Belts} = \frac{\text{Design Power}}{\text{Power Per Belt}} = \frac{3.6554}{1.67} = 2.188 \approx 3 \text{ belts}$$

Coupling Design

1. Input Power= 3.5 HP
2. Input Speed = 750 rpm

1- Select a type of coupling

Choose a cast iron rigid coupling → Cast iron

2- Selecting a proper material with a yield strength from the table material

From table of (AISI) 1030 **asrolled** with yield strength (σ yield) 344.7 MPa, Factor of safety= 6

Permissible shear stress for bolts, shaft, and key material: $\frac{0.5 S_Y}{F.S} = \frac{0.5 \cdot 344.7}{6} = 28.725 \text{ MPa}$

Crushing stress for bolt and key material:

$$\frac{S_Y}{F.S} = \frac{344.7}{6} = 57.45 \text{ MPa}$$

Permissible shear stress for cast iron: **8 MPa** (assume)

3- Get Torque from power

$$\text{Power} = 3.5 \text{ Hp} = 2611 \text{ W} = 2.611 \text{ KW}$$

$$\text{Power} = T \omega$$

$$T = \frac{\text{input Power}}{\text{input speed}} = \frac{P}{\frac{2\pi N}{60}} = \frac{2.611 \cdot 10^3}{\frac{2\pi \cdot 750}{60}} = 33.25 \text{ N.m}$$

Mechanical Properties of Selected Carbon and Alloy Steels

AISI Number	Treatment	Tensile Strength		Yield Strength		Elongation (%)	Reduction in Area (%)	Brinell Hardness, H_B	Izod Impact Strength	
		MPa	ksi	MPa	ksi				J	ft.-lb
1015	As-rolled	420.6	61.0	313.7	45.5	39.0	61.0	126	110.5	81.5
	Normalized	424.0	61.5	324.1	47.0	37.0	69.6	121	115.5	85.2
	Annealed	386.1	56.0	284.4	41.3	37.0	69.7	111	115.0	84.8
1020	As-rolled	448.2	65.0	330.9	48.0	36.0	59.0	143	86.8	64.0
	Normalized	441.3	64.0	346.5	50.3	35.8	67.9	131	117.7	86.8
	Annealed	394.7	57.3	294.8	42.8	36.5	66.0	111	123.4	91.0
1030	As-rolled	551.6	80.0	344.7	50.0	32.0	57.0	179	74.6	55.0
	Normalized	520.6	75.5	344.7	50.0	32.0	60.8	149	93.6	69.0
	Annealed	463.7	67.3	341.3	49.5	31.2	57.9	126	69.4	51.2
1040	As-rolled	620.5	90.0	413.7	60.0	25.0	50.0	201	48.8	36.0
	Normalized	589.5	85.5	374.0	54.3	28.0	54.9	170	65.1	48.0
	Annealed	518.8	75.3	353.4	51.3	30.2	57.2	149	44.3	32.7
1050	As-rolled	723.9	105.0	413.7	60.0	20.0	40.0	229	31.2	23.0
	Normalized	748.1	108.5	427.5	62.0	20.0	39.4	217	27.1	20.0
	Annealed	636.0	92.3	365.4	53.0	23.7	39.9	187	16.9	12.5
1095	As-rolled	965.3	140.0	572.3	83.0	9.0	18.0	293	4.1	3.0
	Normalized	1013.5	147.0	499.9	72.5	9.5	13.5	293	5.4	4.0
	Annealed	656.7	95.3	379.2	55.0	13.0	20.6	192	2.7	2.0

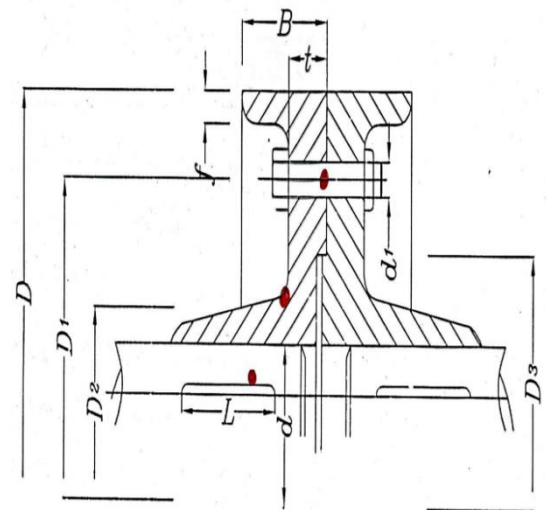
5- Coupling dimensions proportions (see Figure) for the nomenclature

1. No. of bolts $= n = 0.02d + 3 = 0.02 \cdot 20 + 3 = 3.4 \approx$
4 bolts (even number for symmetry)
2. $d_1(\text{diameter of bolts}) = \frac{0.5d}{\sqrt{n}} = \frac{0.5 \cdot 20}{\sqrt{4}} = 5 \text{ mm} \approx$ **M6**
3. $D_1(\text{diameter of pitch circle}) = 4d = 4 \cdot 20 =$ **80 mm**
4. $D_2(\text{outer diameter of hub}) = 2d = 2 \cdot 20 =$ **40 mm**
5. $L_1(\text{length of hub}) = 1.5d = 1.5 \cdot 20 =$ **30 mm**
6. $D(\text{outside diameter of flange}) = 5d = 5 \cdot 20 =$ **100 mm**
(Diameter of Coupling)
7. $t = 1.2 d_1 = 1.2 \cdot 6 =$ **7.2 mm**
8. $f =$ **6 for Cast Iron**

From table of Metric standard keyway and key size at $d = 20 \text{ mm}$

Keyway: **W = 6 mm, h = 2.8 mm**

Key: **W = 6 mm, h = 6 mm**



Metric Standard Parallel Keyway and Key Sizes

Shaft Diameter (mm)		Keyway (mm)		Key (mm)*	
From	To	Width (W)	Depth (h)	Width (W)	Depth (T)
6	8	2	1.0	2	2
9	10	3	1.4	3	3
11	12	4	1.8	4	4
13	17	5	2.3	5	5
18	22	6	2.8	6	6
23	30	8	3.3	8	7
31	38	10	3.3	10	8
39	44	12	3.3	12	8
45	50	14	3.8	14	9

6- the following checks should be done after the assumptions of the coupling dimensions

1- Direct shear check on bolts:

$$F_{bolt} = \frac{2T}{nD_1} = \frac{2 \cdot 33,25}{4 \cdot 80 \cdot 10^{-3}} = 207,8 \text{ N}$$

$$\tau_{bolt} = \frac{F}{\frac{\pi}{4} d_1^2} = \frac{207,8}{\frac{\pi}{4} \cdot 6^2} = 7,36 \text{ MPa} < 28.7 \text{ MPa}$$

∴ It's SAFE

2- Bearing pressure on bolts:

$$\sigma_b = \frac{F_b}{d_1 \cdot t} = \frac{207,8}{6 \cdot 7.2} = 4.81 \text{ MPa}$$

$$\sigma_b = \frac{S_b}{F.S} \rightarrow 4.81 < \frac{57.45 \times 10^6}{6} = 9.575 \text{ MPa}$$

∴ It's SAFE

3- Shear of the flange at hub:

$$\tau = \frac{2T}{\pi D_2^2 t} = \frac{2 \cdot 33.25}{\pi \cdot 40^2 \cdot 7.2} = 1.84 \text{ MPa} < 8 \text{ MPa}$$

∴ It's SAFE

4-Shear Stress on Key

$$\tau_{key} = \frac{2T}{d \cdot w \cdot L} = \frac{2 \cdot 33.25}{20 \cdot 6 \cdot 30} = 18.4 \text{ MPa} < 28.725 \text{ MPa}$$

∴ It's SAFE

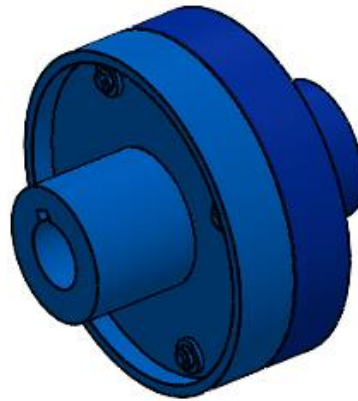
5- on Key

$$\sigma_b =$$

$$\frac{2 \cdot 33.25}{20 \cdot \frac{6}{2} \cdot 30} =$$

$$36.9 \text{ MPa} < 57.45 \text{ MPa}$$

∴ It's SAFE



Crushing

$$\frac{2T}{d \cdot \frac{h}{2} \cdot L} =$$

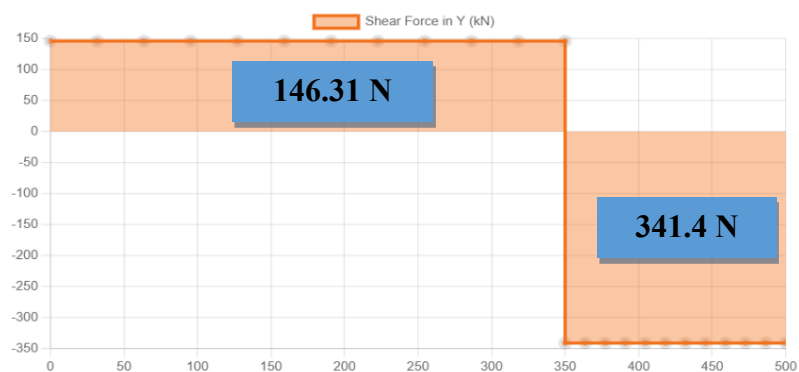
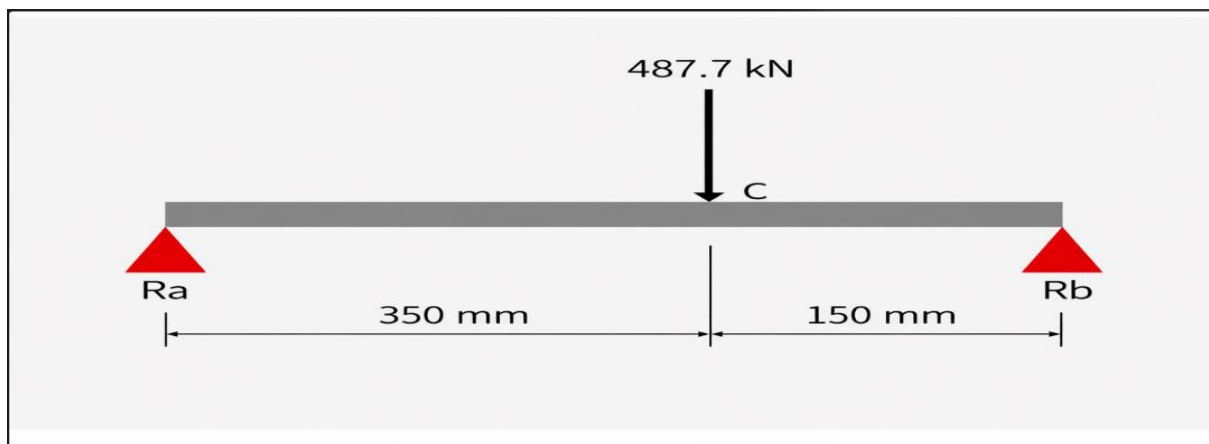
Shafts Design

1st Shaft: (Helical)

Power = 2.611 kW

(X-Y Plane)

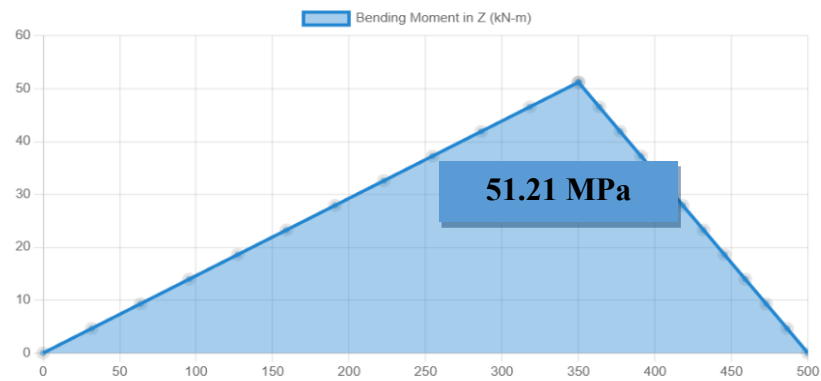
- ❖ Tangent force $w_t = 1340 \text{ N}$
- ❖ Axial force $F_a = w_t \tan \beta = 487.7 \text{ N}$
- ❖ Normal force $F_N = w_t \tan \alpha = 487.7 \text{ N}$
- ❖ $R_A = 146.31 \text{ N}$
- ❖ $R_B = 341.4 \text{ N}$



(X-Z
Plane)

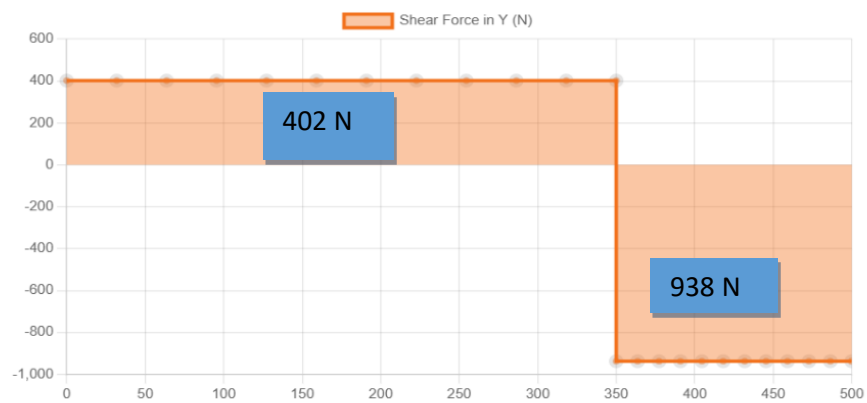
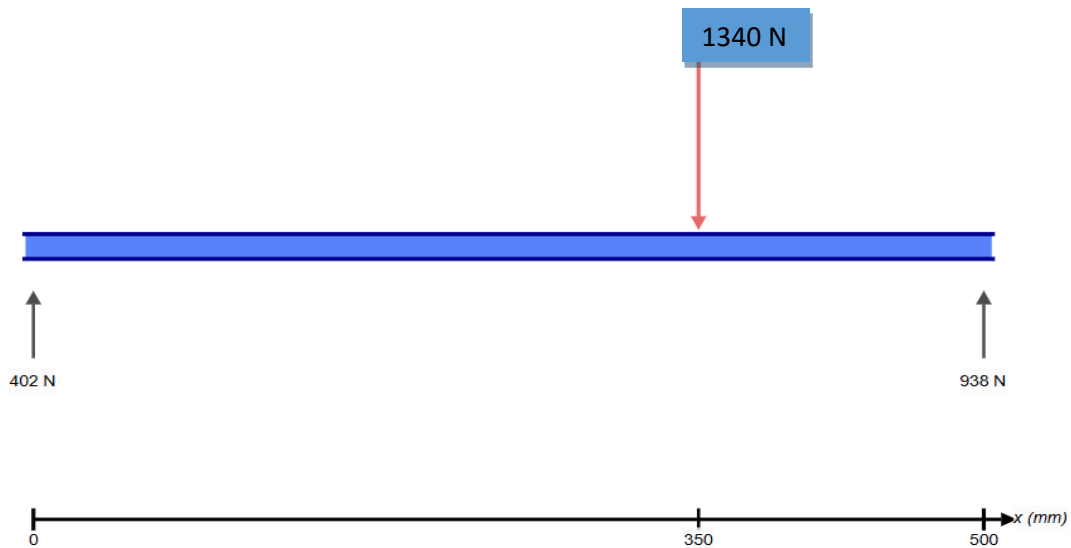
$R_A=938$

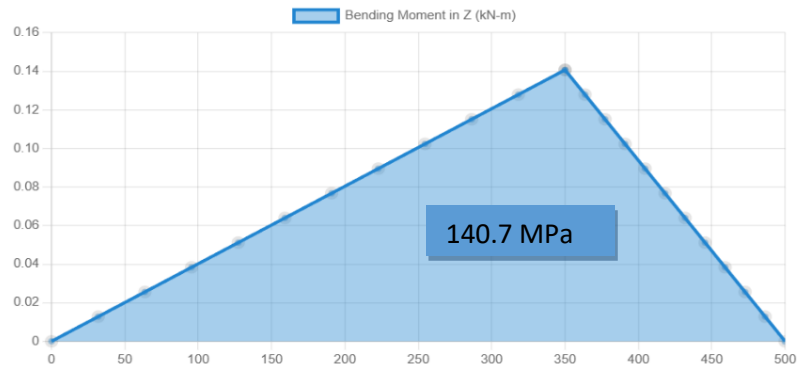
$R_B=402$



N

N





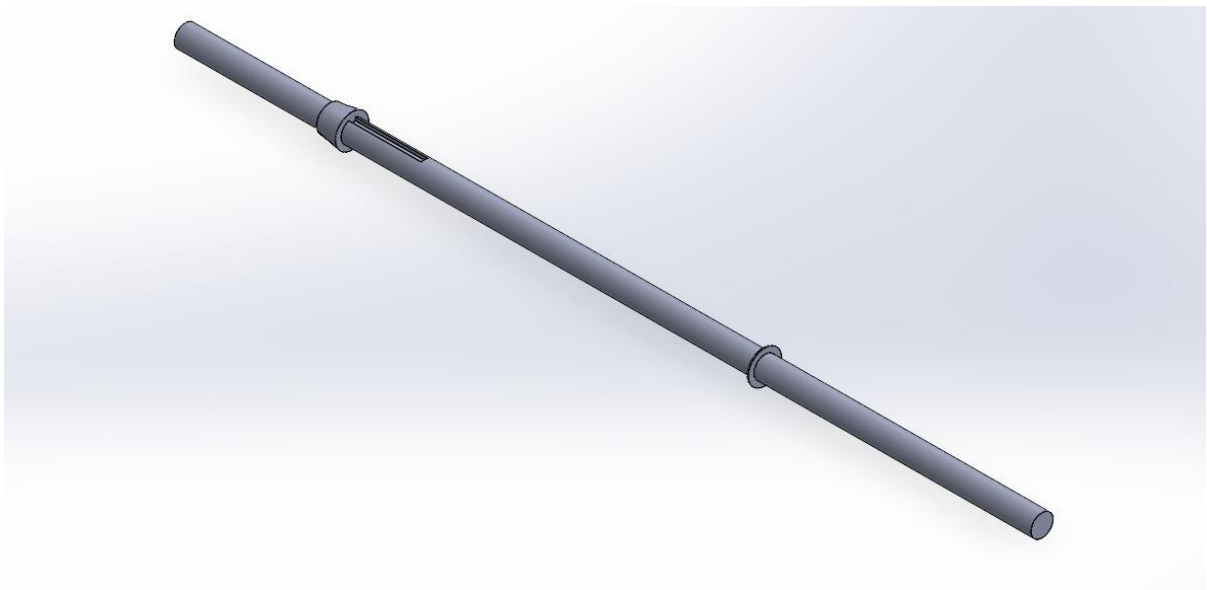
$$Moment_{max} = \sqrt{140.7^2 + 51.21^2 + 12.071^2} = 150.21 \text{ Nm}$$

$$Torque_{max} = 33.24 \text{ Nm}$$

$$K_m = K_t = 1.5$$

$$d^3 = \frac{32 * FS}{\pi * S_y} * \sqrt{(K_t * Torque_{max})^2 + (K_m * Moment_{max})^2}$$

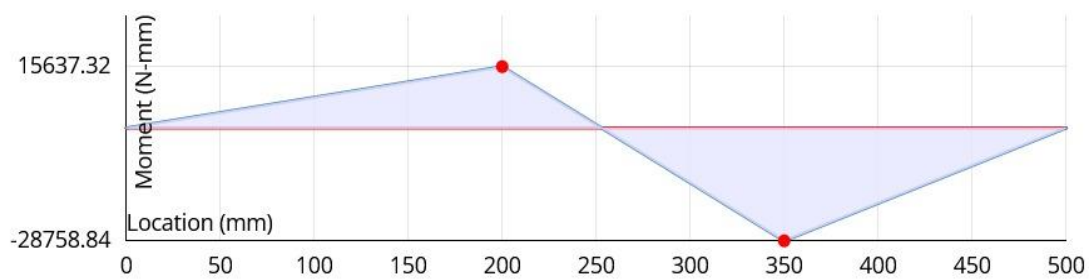
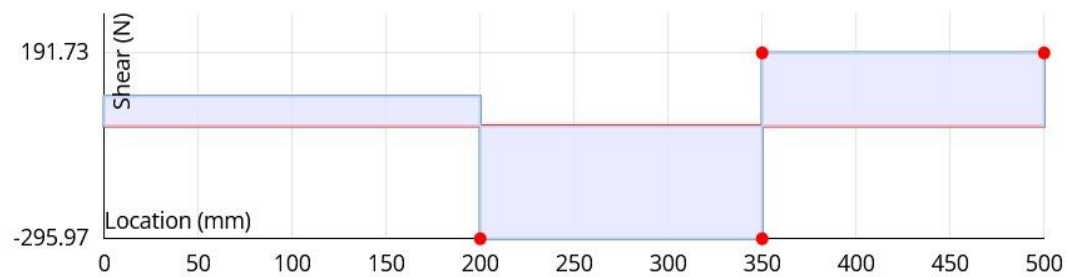
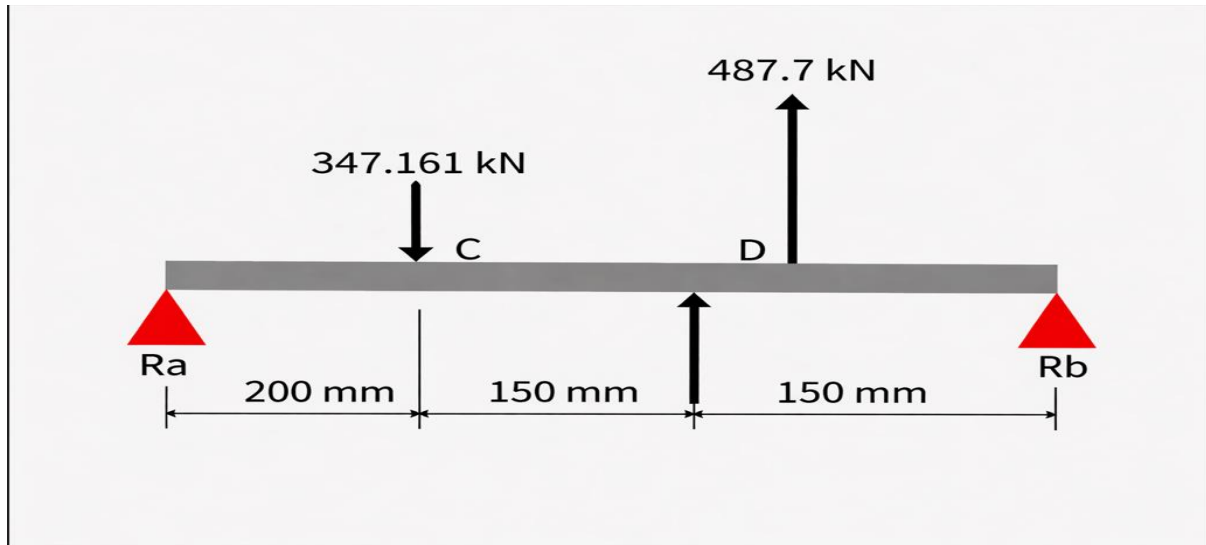
$$d = 18.3 \text{ mm} \rightarrow \text{Choose } 20 \text{ mm}$$



2nd Shaft: (Spur + Helical)

Power = 2.611 kW

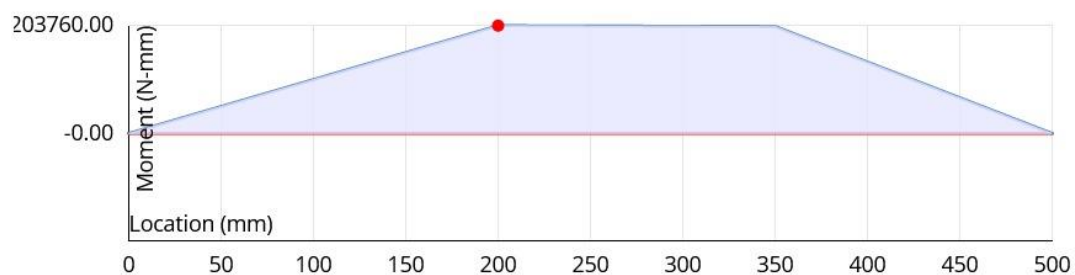
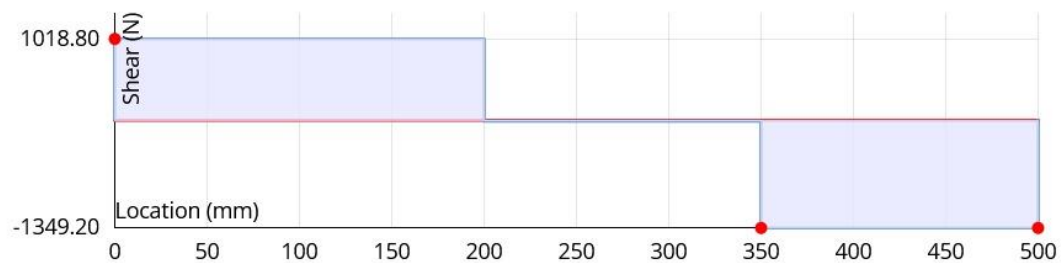
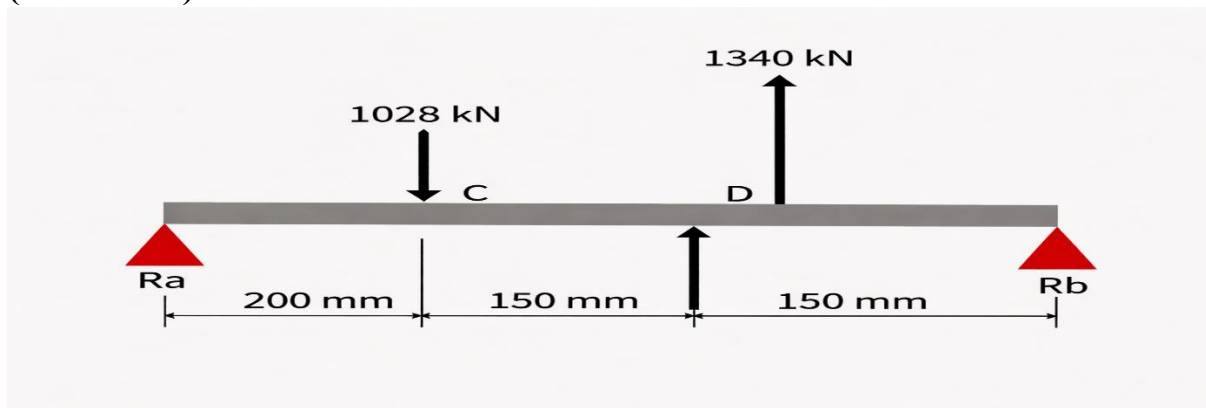
(X-Y Plane)



$$R_A = 78.181 \text{ N}$$

$$R_B = 191.72 \text{ N}$$

(X-Z Plane)



$$R_A = 1018.8 \text{ N}$$

$$R_B = 1349.2 \text{ N}$$

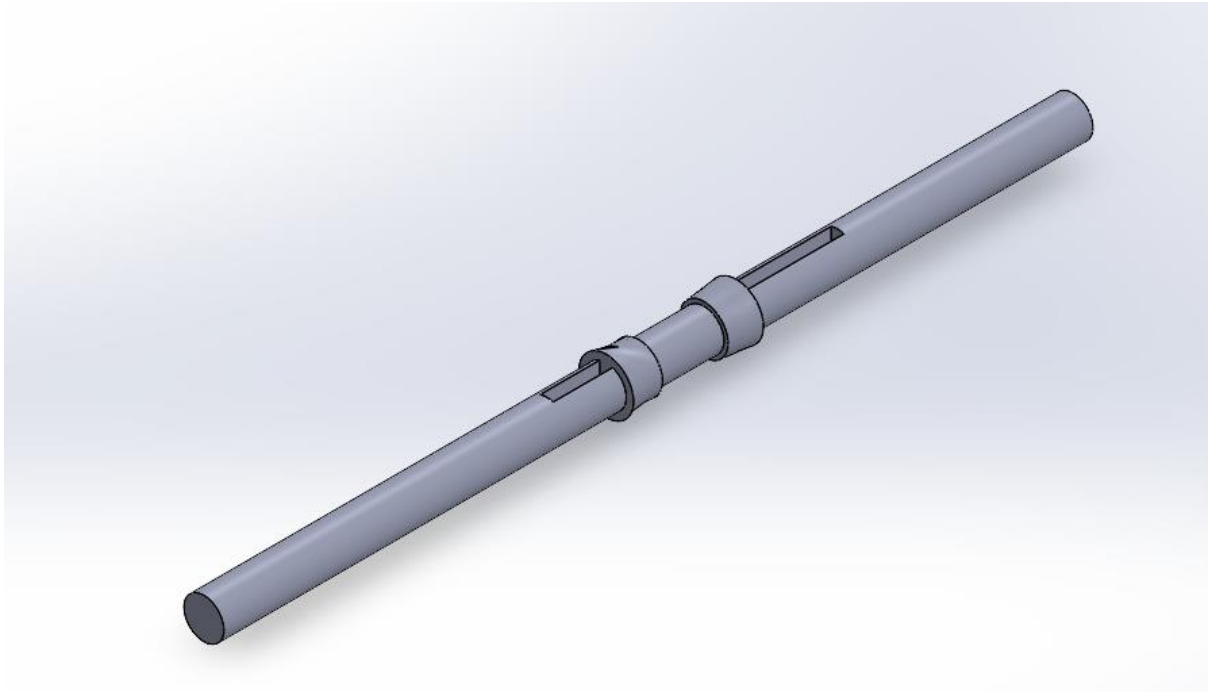
$$Moment_{max} = 150.215 \text{ Nm}$$

$$Torque_{max} = 33.24 \text{ Nm}$$

$$K_m = K_t = 1.5$$

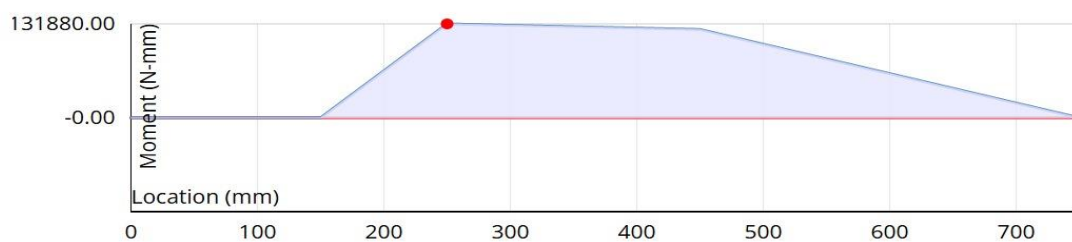
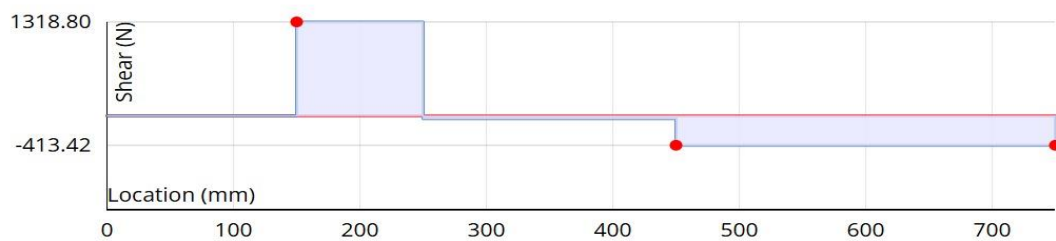
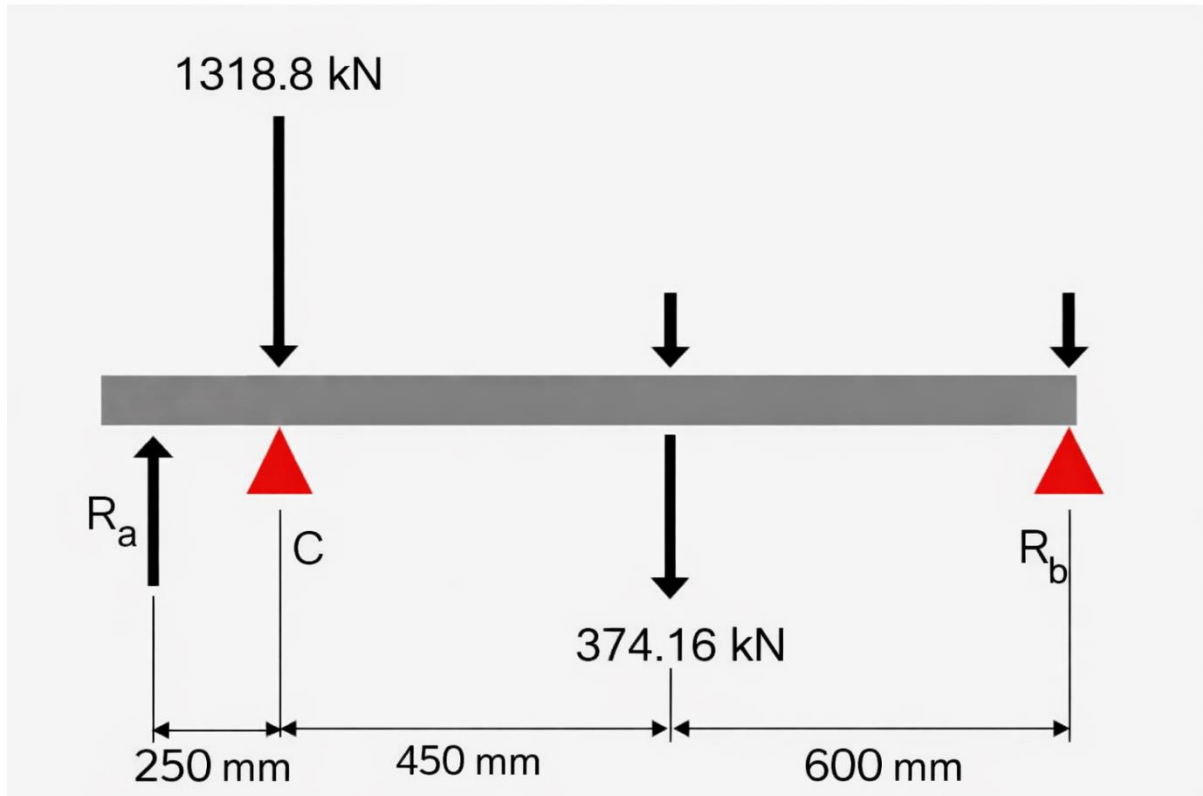
$$d^3 = \frac{32 * FS}{\pi * S_y} * \sqrt{(K_t * Torque_{max})^2 + (K_m * Moment_{max})^2}$$

$$d = 22.5 \rightarrow \text{Choose } 25 \text{ mm}$$



3rd Shaft: (Helical + Belt Pulley)

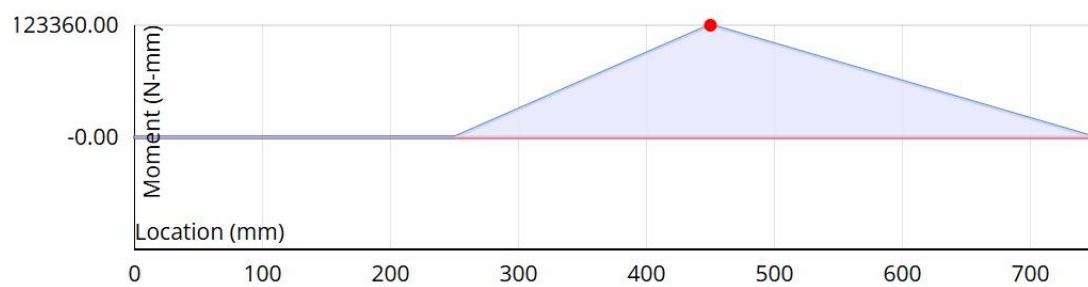
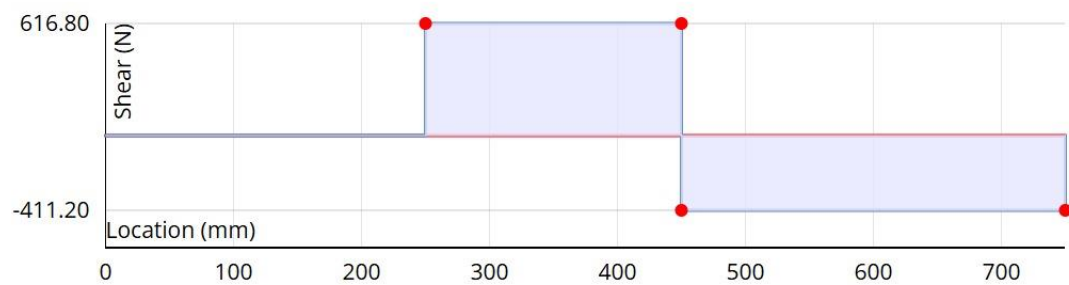
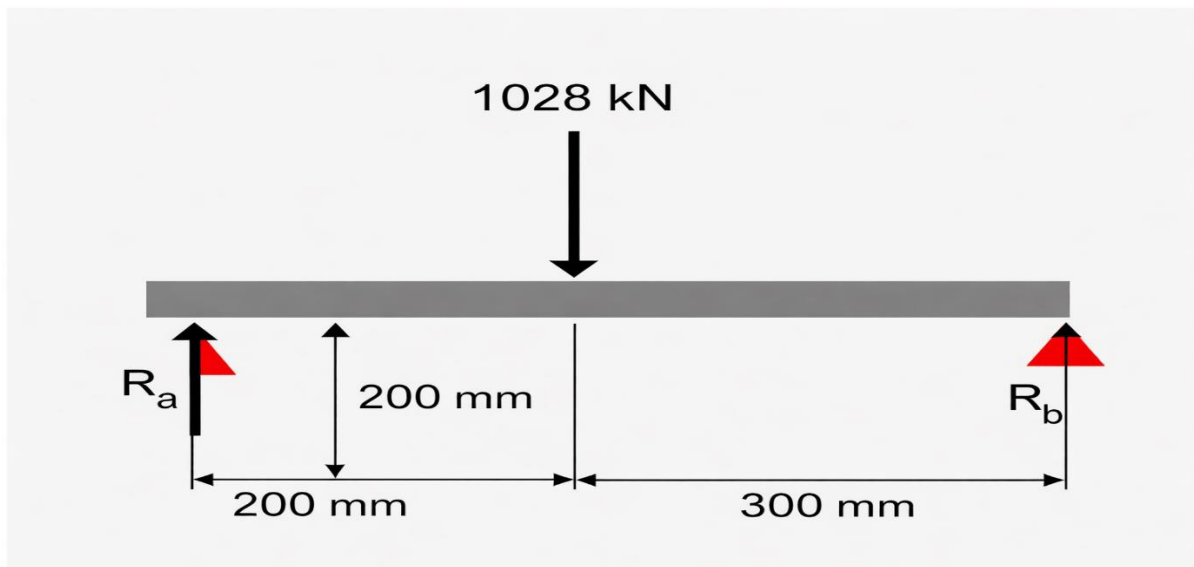
(X-Y Plane)



$$R_A = 1358.064 \text{ N}$$

$$R_B = 413.405 \text{ N}$$

(X-Z Plane)



$$R_A = 616.8 \text{ N}$$

$$R_B = 411.2 \text{ N}$$

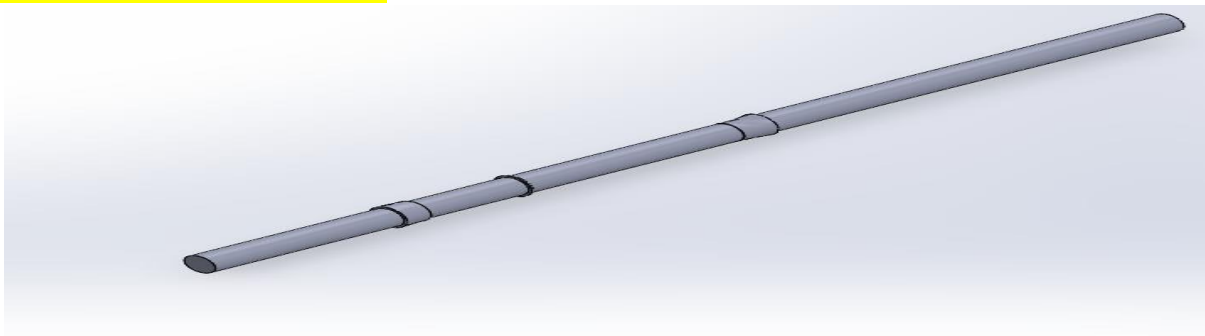
$$Moment_{max} = 180.58 \text{ Nm}$$

$$Torque_{max} = 19.78 \text{ Nm}$$

$$K_m = K_t = 1.5$$

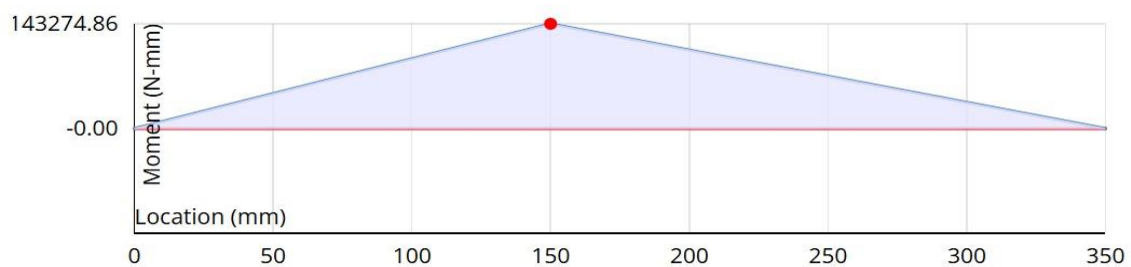
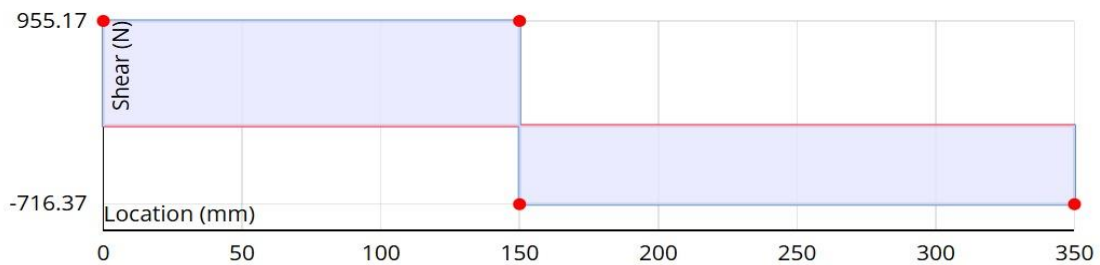
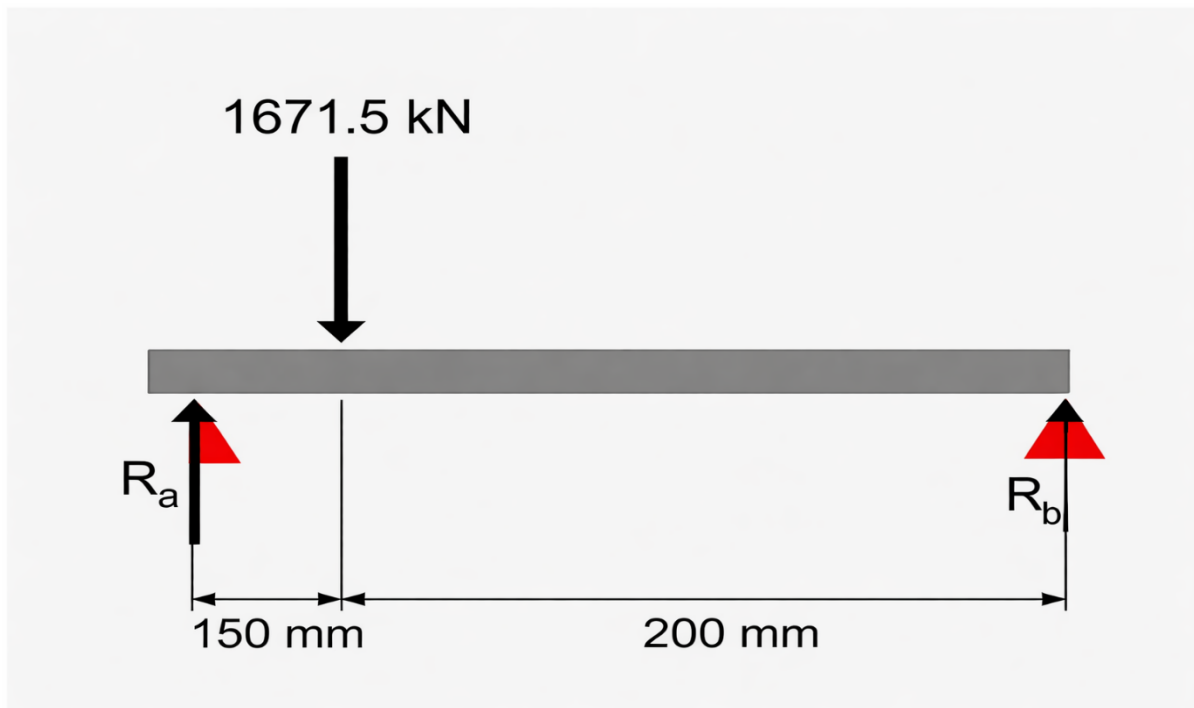
$$d^3 = \frac{32 * FS}{\pi * S_y} * \sqrt{(K_t * Torque_{max})^2 + (K_m * Moment_{max})^2}$$

$$d = 21.5 \rightarrow \text{Choose } 25\text{mm}$$



4th Shaft: (Belt Pulley)

(X-Y Plane)



$$R_A = 955.17 \text{ N}$$

$$R_B = 716.378 \text{ N}$$

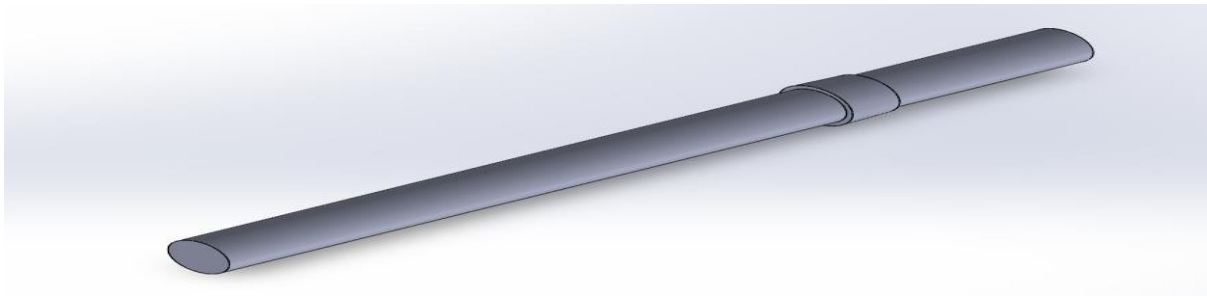
$$Moment_{max} = 143,27 \text{ Nm}$$

$$Torque_{max} = 1671.549 \text{ Nm}$$

$$K_m = K_t = 1.5$$

$$d^3 = \frac{32 * FS}{\pi * S_y} * \sqrt{(K_t * Torque_{max})^2 + (K_m * Moment_{max})^2}$$

$$d = 20 \rightarrow \text{Choose } 25\text{mm}$$



Bearings Design

1st Shaft:

Choose ball bearing

1- Select bearing type from the data sheet table, then get C, C₀

At diameter 20 mm

RPm = 750

Fr = 998.95 KN

Fa = 773.65 KN

From table bearing type: Single row deep groove ball bearings 61804-2RS1

Principal dimensions			Basic load ratings		Fatigue load limit	Speed ratings		Mass	Designations	
d	D	B	C	C ₀	P _u	Reference speed	Limiting speed ¹⁾		Bearing open or capped on both sides	capped on one side ¹⁾
mm			kN		kN	r/min		kg	–	
17 cont.	47	14	14,3	6,55	0,275	34 000	22 000	0,11	► 6303	–
	47	14	14,3	6,55	0,275	–	11 000	0,12	► 6303-2RSH	6303-RSH
	47	14	14,3	6,55	0,275	34 000	17 000	0,12	6303-2RSL	6303-RSL
	47	14	14,3	6,55	0,275	34 000	17 000	0,12	► 6303-2Z	6303-Z
	47	19	13,5	6,55	0,275	–	11 000	0,16	62303-2RS1	–
	62	17	22,9	10,8	0,455	28 000	18 000	0,27	6403	–
20	32	7	4,03	2,32	0,104	–	13 000	0,018	► 61804-2RS1	–
	32	7	4,03	2,32	0,104	45 000	22 000	0,018	► 61804-2RZ	–
	32	7	4,03	2,32	0,104	45 000	28 000	0,018	► 61804	–
	37	9	6,37	3,65	0,156	–	12 000	0,038	► 61904-2RS1	–
	37	9	6,37	3,65	0,156	43 000	20 000	0,038	► 61904-2RZ	–
	37	9	6,37	3,65	0,156	43 000	26 000	0,037	► 61904	–

C = 4.03 kN, C₀ = 2.32 kN

$$\Sigma F_y |_B = \sqrt{(341.4)^2 + (938)^2} = 998.095 \text{ N}$$

$$F_R = 998.95 \text{ N}$$

$$F_a = F_R \tan (20^\circ) = 1340 \tan (20^\circ) = 773.65 \text{ N}$$

2- Get P

$$\therefore P = 0.56F_r + yF_a \quad @ \quad y = \frac{0.44}{e}$$

$$\therefore e = 0.5 \left(\frac{F_a}{C_0} \right)^{0.225} = 0.39$$

$$\therefore \text{If } \frac{F_a}{F_r} \leq e \text{ then } P = F_r$$

$$\text{If } \frac{F_a}{F_r} \geq e \text{ then } P = 0.56F_r + yF_a$$

$$\therefore \frac{F_a}{F_r} \geq e$$

$$0.774 \geq 0.39$$

$$\therefore P = 0.56 \cdot 998.95 + y \cdot 773.652 @ y = \frac{0.44}{e} = 1.128$$

$$P = 1432.294 \text{ N}$$

3-Assume life, for gear box L=12,000 working hr

4- Calculate $L = \frac{\left[\frac{C}{P}\right]^a \times 10^6}{N \cdot 60}$ to get C where a=3 (ball bearing)

$$L = \frac{\left[\frac{C}{P}\right]^a \times 10^6}{N \cdot 60}; 12,000 = \frac{\left[\frac{C}{1432.294}\right]^3 \times 10^6}{* 60}$$

From equation above, get C = 11.663 kN > 4.03 kN

IT'S NOT SAFE

$\therefore$ Change material to 6204

$$C = 13.55 \text{ kN } C_o = 6.55 \text{ kN}$$

$$\therefore e = 1.463y = \frac{0.44}{e} = 0.3$$

- $P = F_r = 998.65 \text{ N}$

- $\frac{\left[\frac{C}{P}\right]^a \times 10^6}{N \cdot 60} C = 8.132 \text{ kN}$

- $C < C \quad 8.132 < 13.55$

$\therefore$ IT'S SAFE

2nd Shaft:

Given data:

$$d_{shaft} = 25 \text{ mm}$$

$$N = 970 \text{ rpm}$$

$$F_a = 773,65 \text{ N}$$

$$F_r = 1362 \text{ N}$$

Choose ball bearing

1- Select bearing type from the data sheet table, then get C , C_o

Principal dimensions			Basic load ratings		Fatigue load limit	Speed ratings		Mass	Designations	
d	D	B	C	C_o	P_u	Reference speed	Limiting speed ¹⁾		Bearing open or capped on both sides	capped on one side ¹⁾
mm			kN		kN	r/min		kg	–	
25	37	7	4,36	2,6	0,125	–	11 000	0,022	► 61805-2RS1	–
	37	7	4,36	2,6	0,125	38 000	19 000	0,022	► 61805-2RZ	–
	37	7	4,36	2,6	0,125	38 000	24 000	0,022	► 61805	–
	42	9	7,02	4,3	0,193	–	10 000	0,045	► 61905-2RS1	–
	42	9	7,02	4,3	0,193	36 000	18 000	0,045	► 61905-2RZ	–
	42	9	7,02	4,3	0,193	36 000	22 000	0,045	► 61905	–
	47	8	8,06	4,75	0,212	32 000	20 000	0,055	► 16005	–
	47	12	11,9	6,55	0,275	32 000	20 000	0,078	► 6005	–
	47	12	11,9	6,55	0,275	–	9 500	0,081	► 6005-2RSH	6005-RSH
	47	12	11,9	6,55	0,275	32 000	16 000	0,08	► 6005-2RSL	6005-RSL
	47	12	11,9	6,55	0,275	32 000	16 000	0,083	► 6005-2Z	6005-Z
	47	16	11,2	6,55	0,275	–	9 500	0,11	63005-2RS1	–
	52	15	14,8	7,8	0,335	28 000	18 000	0,13	► 6205	–
	52	15	14,8	7,8	0,335	–	8 500	0,13	► 6205-2RSH	6205-RSH
	52	15	14,8	7,8	0,335	28 000	14 000	0,13	► 6205-2RSL	6205-RSL

At diameter 25 mm

From table bearing type: **6205-2RSH**

$$C = 14.8 \text{ kN}, C_o = 7.8 \text{ kN}$$

$$F_R \mid_B = \sqrt{(1352.2)^2 + (191.78)^2} = 1362 \text{ N}$$

$$F_a = F_R \tan (30^\circ)$$

$$F_a = 1340 \tan (30^\circ) = 773.65 \text{ N}$$

2-Assume life, for gear box L=12,000 working hr

3-Get P

$$e = 0.5 \left(\frac{F_a}{C_o} \right)^{0.225} = 0.5 \left(\frac{773,65}{7800} \right)^{0.225} = 0.2973$$

$$\frac{F_a}{F_r} = \frac{773,65}{1362} = 0.568; > e, \text{ then } P = 0.56 F_r + \left(\frac{0.44}{e} \right) F_a$$

$$P = 0.56 * 1362 + \left(\frac{0.44}{0.2973} \right) * 773,65 = 1907.722 \text{ N}$$

4- Calculate $L = \frac{\left[\frac{C}{P} \right]^a \times 10^6}{N * 60}$ to get C where $a=3$ (ball bearing)

$$L = \frac{\left[\frac{C}{P} \right]^a \times 10^6}{N * 60} = 12,000 = \frac{\left[\frac{C}{1907.722} \right]^3 \times 10^6}{970 * 60}$$

From the equation above, get $C' = 16.925 \text{ kN} > 14.8 \text{ kN}$

IT'S NOT SAFE

- Changing material to 6305 ETN9

$$C = 26 \text{ kN } C_o = 13.4 \text{ kN}$$

$$e = 0.2632 \quad y = \frac{0.44}{e} = 1.6717$$

$$P = 0.56 * 1362 + 1.6717 * 773.65 = 2056.03 \text{ N}$$

$$C' = 18.241 \text{ kN}$$

$$\therefore C' < C$$

$$18.241 < 26$$

IT'S SAFE

3rd Shaft:

Given data:

$$d_{shaft} = 25\text{mm}$$

$$N = 1260 \text{ rpm}$$

$$F_a = 0$$

$$F_r = 1491.51\text{N}$$

$$17.8 \text{ 9.3}$$

Choose ball bearing

1- Select bearing type from the data sheet table, then get C , C_0

At 25 mm diameter

Principal dimensions			Basic load ratings		Fatigue load limit	Speed ratings		Mass	Designations	
d	D	B	C	C ₀	P _u	Reference speed	Limiting speed ¹⁾		Bearing open or capped on both sides	capped on one side ¹⁾
mm			kN		kN	r/min		kg	–	
25	37	7	4,36	2,6	0,125	–	11 000	0,022	► 61805-2RS1	–
	37	7	4,36	2,6	0,125	38 000	19 000	0,022	► 61805-2RZ	–
	37	7	4,36	2,6	0,125	38 000	24 000	0,022	► 61805	–
	42	9	7,02	4,3	0,193	–	10 000	0,045	► 61905-2RS1	–
	42	9	7,02	4,3	0,193	36 000	18 000	0,045	► 61905-2RZ	–
	42	9	7,02	4,3	0,193	36 000	22 000	0,045	► 61905	–
	47	8	8,06	4,75	0,212	32 000	20 000	0,055	► 16005	–
	47	12	11,9	6,55	0,275	32 000	20 000	0,078	► 6005	–
	47	12	11,9	6,55	0,275	–	9 500	0,081	► 6005-2RSH	6005-RSH
	47	12	11,9	6,55	0,275	32 000	16 000	0,08	► 6005-2RSL	6005-RSL
	47	12	11,9	6,55	0,275	32 000	16 000	0,083	► 6005-2Z	6005-Z
	47	16	11,2	6,55	0,275	–	9 500	0,11	63005-2RS1	–
	52	15	14,8	7,8	0,335	28 000	18 000	0,13	► 6205	–
	52	15	14,8	7,8	0,335	–	8 500	0,13	► 6205-2RSH	6205-RSH
	52	15	14,8	7,8	0,335	28 000	14 000	0,13	► 6205-2RSL	6205-RSL
	52	15	14,8	7,8	0,335	28 000	14 000	0,13	► 6205-2Z	6205-Z
	52	15	17,8	9,3	0,4	28 000	18 000	0,12	6205 ETN9	–
	52	18	14	7,8	0,335	–	8 500	0,13	62205-2RS1	–
	62	17	23,4	11,6	0,49	24 000	16 000	0,23	► 6305	–
	62	17	23,4	11,6	0,49	–	7 500	0,24	► 6305-2RSH	6305-RSH
	62	17	23,4	11,6	0,49	24 000	13 000	0,23	6305-2RZ	6305-RZ

From table bearing type: Single row deep groove ball bearings 6205 ETN9

$$C = 17.8 \text{ kN}, C_0 = 9.3 \text{ kN}$$

$$F_R|_A = \sqrt{(1358)^2 + (616.8)^2} = 1491.51 \text{ N}$$

$$F_R|_B = \sqrt{(411.75)^2 + (415.4)^2} = 583 \text{ N}$$

2-Assume life, for gear box L=12,000 working hr

3-Get P

$$P = F_r = 1491.5 \text{ N}$$

4- Calculate $L = \frac{\left[\frac{C}{P}\right]^a \times 10^6}{N \times 60}$ to get C where a=3 (ball bearing)

$$L = \frac{\left[\frac{C}{P}\right]^a \times 10^6}{N \times 60} = 12,000 = \frac{\left[\frac{C}{1491.51}\right]^3 \times 10^6}{1260 \times 60}$$

From the equation above, get C' = 14.438kN < 17.8 kN

IT'S SAFE

4th Shaft:

Given data:

$$d_{shaft} = 25 \text{ mm}$$

$$N = 1600$$

$$F_a = 0$$

$$F_r = 955.17 \text{ N}$$

Choose ball bearing

1- Select bearing type from the data sheet table, then get C, C_o

At 25 mm diameter

From table bearing type: Single row deep groove ball bearings 6005-2RSH

$$C = 11.9 \text{ kN}, C_o = 6.55 \text{ kN}$$

$$F_R|_A = 955.17 \text{ N}$$

$$F_R|_B = 716.378 \text{ N}$$

2-Assume life, for gear box L=12,000 working hr

3-Get P

$$\therefore F_a = 0$$

$$\therefore P = F_r = 955.17 \text{ N}$$

4- Calculate $L = \frac{\left[\frac{C}{P}\right]^a \times 10^6}{N \cdot 60}$ to get C where a=3 (ball bearing)

$$L = \frac{\left[\frac{C}{P}\right]^a \times 10^6}{N \cdot 60} = 12,000 = \frac{\left[\frac{C}{955.17}\right]^3 \times 10^6}{300 \cdot 60}$$

From the equation above, get C' = 10.013 kN < 11.9 kN

IT'S SAFE

Assembly Drawings (SolidWorks)

